

Abdul Rahman Shaikh

☎ +1-224-848-2725 ✉ iamsabdulrahman@gmail.com 📍 Chicago, IL 🏠 sabdulrahman.github.io
🌐 linkedin.com/in/iamsabdulrahman 🐙 github.com/sabdulrahman 🎓 [Google Scholar](#)

AREA OF EXPERTISE

Ph.D. candidate in Computer Science specializing in artificial intelligence, machine learning, large language models, and scalable AI systems. My research focuses on intelligent systems for complex, multimodal, and high-volume data. Experienced in LLMs, retrieval-augmented generation, multimodal learning, multi-agent systems, and AI evaluation. Led research projects, mentored student researchers, and published 10+ papers in venues such as IEEE VIS and TVCG.

EDUCATION

Northern Illinois University <i>Ph.D. Computer Science (GPA: 3.9)</i>	Dekalb, IL <i>Aug 2020 – Present</i>
Northern Illinois University <i>M.S. Computer Science (GPA: 3.9)</i> <i>Thesis: Modeling the Broader Impact of Science and Health Using Social Media</i>	Dekalb, IL <i>Jan 2018 – May 2020</i>
Osmania University <i>B.E. Computer Science (GPA: 3.6)</i>	Hyderabad, India <i>Aug 2013 – Aug 2017</i>

EXPERIENCE

AI/ML Researcher (DATA Lab, VA Lab & WASTE Lab) <i>Northern Illinois University</i>	Dekalb, IL <i>Jan 2018 – Present</i>
- Conducted AI, ML, and LLM research across three labs, developing pipelines for 1M+ records to support large-scale experimentation and multimodal analysis. - Led Visual Analytics Lab research initiatives, mentoring 5+ graduate student researchers through structured milestones, code reviews, and experiment design, contributing to 4 peer-reviewed publications. - Designed and evaluated an LLM-assisted cross-view visualization framework, reducing data exploration time by 35% in controlled user studies with 25+ participants. - Designed reproducible evaluation frameworks for LLM and ML systems using benchmarking, ablation studies, error analysis, Ragas, TruLens, and LLM-as-judge methods. - Built multi-agent research workflows using LangChain, AutoGen, and CrewAI for literature synthesis, retrieval, and automated analysis, reducing review time by 50%+. - Developed GenAI workflows for captioning, reasoning, and topic discovery across 300K+ social media images, supporting research on public-health communication and visual content analysis. - Published 10+ peer-reviewed papers in IEEE VIS, IEEE TVCG, and JCDL, with contributions spanning ML, AI systems, visual analytics, and data-driven decision support.	
Teaching Assistant (Algorithms, Databases, C/C++, & Java) <i>Northern Illinois University</i>	Dekalb, IL <i>Aug 2018 – May 2023</i>
- Supported instruction for 70+ students across algorithms, databases, C/C++, and Java, reinforcing concepts in data structures, memory management, SQL, and problem solving. - Evaluated programming assignments, database projects, and algorithmic exercises, providing feedback on correctness, efficiency, code quality, and query optimization. - Led weekly lab sessions, office hours, and individualized mentoring to strengthen student understanding of programming fundamentals and computational thinking. - Assisted with course preparation, grading rubrics, and instructional materials, helping align assignments and labs with course learning objectives.	

AI/ML Engineer

Visual Technologies

Remote

Aug 2024 – Present

- Built Python-based AI/ML pipelines for multimodal datasets using PyTorch, Hugging Face, OpenCV, Pandas, and Docker, supporting scalable experimentation and model evaluation.
- Developed RAG systems using LangChain, Hugging Face Transformers, OpenAI API, FAISS, and PostgreSQL, improving retrieval quality on unstructured document collections.
- Trained deep learning models for visual classification, feature extraction, and semantic search using PyTorch, CLIP, and Sentence Transformers.
- Implemented embedding generation, vector indexing, and metadata-filtered search across text and image datasets, reducing information lookup time from hours to minutes.
- Built AI evaluation workflows using MLflow, Weights & Biases, and custom test sets to track accuracy, latency, hallucination rate, and retrieval quality across deployments.

Data Analyst

Amazon

Hyderabad, India

Aug 2016 – Dec 2017

- Built SQL and MongoDB data pipelines to analyze operational datasets, identify performance bottlenecks, and reduce query latency by 25%.
- Used Python, Pandas, NumPy, and Scikit-learn for data cleaning, exploratory analysis, and quality assurance, improving reliability of reporting datasets.
- Collaborated with engineering and operations teams to translate data analysis into actionable insights for database performance and workflow improvement.

Software Engineer

BotHook

Hyderabad, India

Jan 2015 – Aug 2016

- Built rule-based and NLP-driven chatbots for client websites and messaging platforms, automating customer support, lead capture, and FAQ workflows.
- Developed backend services and integrations using JavaScript, Node.js, REST APIs, and databases to support real-time chatbot responses and user data retrieval.
- Designed intent flows, dialogue logic, and deployment workflows with startup and client teams, improving response consistency and reducing repetitive support work.

PUBLICATIONS

Published

- **A. R. Shaikh**, M. Sun, X. Liu, H. Alhoori, J. Zhao, and D. Koop, “iTrace: Interactive Tracing of Cross-View Data Relationships”, *Graphics Interface* 2025. [doi: 10.1145/3769872.3769897]
- **A. R. Shaikh**, H. Alhoori, and M. Sun, “YouTube and Science: Models for Research Impact,” *Journal of Scientometrics*, 2022. [doi: 10.1007/s11192-022-04574-5]
- **A. R. Shaikh**, M. Sun, and H. Alhoori, “Toward systematic design considerations of organizing multiple views”, *IEEE Visualization and Visual Analytics (VIS)*, 2022. [doi: 10.1109/VIS54862.2022.00030]
- M. Sun, **A. R. Shaikh**, H. Alhoori, and J. Zhao, “SightBi: Exploring Cross-View Data Relationships with Biclusters,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 28, no. 1, 2022. (Proceedings of IEEE VIS 2021). **Best Paper Honorable Mention**. [doi: 10.1109/TVCG.2021.3114801]
- M. J. Mokarrama, **A. R. Shaikh**, and H. Alhoori, “Examining the Representation of Youth in the US Policy Documents through the Lens of Research,” in *2024 IEEE International Conference on Big Data*, Washington, USA, Dec 15–18, 2024. [doi: 10.1109/BigData62323.2024.10825996]
- M. Rezaei, M. R. Moghadam, **A. R. Shaikh**, H. Alhoori, and R. Freedman, “Generation, Evaluation, and Explanation of Novelists’ Styles with Single-Token Prompts,” *ACM/IEEE Joint Conference on Digital Libraries, JCDL, Dec 15–19, 2025*. [doi:10.48550/arXiv.2511.20459]
- M. Shahzad, H. Alhoori, Reva Freedman, and **A. R. Shaikh**, “Quantifying the online long-term interest in research,” *Journal of Informetrics*, 2022. [doi: 10.1016/j.joi.2022.101288]
- M. Sun, **A. R. Shaikh**, Y. Ma, D. Koop, and H. Alhoori, “Boundary Blending: Reconsidering the Design of Multi-View Visualizations,” 2023. [doi: 10.48550/arXiv.2306.09812]
- **A. R. Shaikh** and H. Alhoori, “Predicting Patent Citations to measure Economic Impact of Scholarly Research,” in *Proceedings of ACM/IEEE Joint Conference on Digital Libraries (JCDL)*, Champaign, Illinois, USA, June 2–6, 2019. [doi: 10.1109/JCDL.2019.00089]
- **A. R. Shaikh**, “Modeling the Broader Impact of Science and Health Using Social Media,” Master’s thesis. [link]

Under Review

- **A. R. Shaikh**, M. Ambati, M. Uddin, and M. Vaezi, “Building Classification: A Comprehensive Dataset and DenseNet201-Based Approach”. *Journal of Urban Planning and Development*
- **A. R. Shaikh**, M. Sun, Y. Ma, D. Koop, and H. Alhoori, “Toward the Design of Transformative Multiple-View Visualizations”. *PacificVis2026*
- M. Karasani, **A. R. Shaikh**, M. Uddin, and M. Vaezi, “Machine-Learning-Based County-Level Prediction of Municipal Solid Waste in the United States with Socioeconomic and Demographic Integration”. *Waste Management & Research: The Journal for a Sustainable Circular Economy (WM&R)*
- N. Vooda, M. Uddin, **A. R. Shaikh**, M. Karasani, M. Hughes and M. Vaezi, “USS-Water Dataset and U-Net+ Model: A Novel High-Resolution Satellite Imagery Approach for Surface Water Detection in the United States”. *PLOS Water Journal*

PROJECTS

- GenHealth: Multimodal Medical Report Analysis** 🔄Code | *Transformers, FastAPI* 2026
- GenHealth is a multimodal AI system that combines text processing, medical imaging analysis, and structured data integration to boost diagnostic extraction accuracy from medical reports.
- Rufus: Intelligent Web Data Extraction for LLMs** 🔄Code | *RAG, Selenium* 2025
- Developed an AI-powered web crawler that extracts and synthesizes relevant content into structured documents for Retrieval-Augmented Generation (RAG) systems.
- LLMFlow: AI-Powered Summarization of Scholarly Documents** 🔄Code | *LangChain* 2025
- Integrated GPT, Claude and Llama with LangChain to generate concise summaries of large-scale research documents, reducing academic reading time.
- MixArt: Generative Artwork with Stable Diffusion** 🔄Code | *Stable Diffusion, LoRA* 2024
- Developed a pipeline for generating abstract and photorealistic art using Stable Diffusion, leveraging LoRA fine-tuning to personalize outputs for user-requested styles.
- VoxCore: Voice Authentication System** 🔄Code | *Python, Whisper, PyTorch* 2024
- Implemented a voice authentication system using OpenAI’s Whisper for transcription and custom PyTorch models for speaker verification, achieving 89% accuracy in multi-speaker environments.
- InstaHealth: Fine-Tuning Caption Generation for Instagram Posts** 🔄Code | *CV* 2023
- Fine-tuned a language model to generate context-aware captions for health-related content on Instagram data, automating large-scale social media captioning.
- Semantic Segmentation with SAM & Custom U-Net** | *Python, CV* 2023
- Integrated the Segment Anything Model (SAM) with a custom U-Net architecture to achieve semantic segmentation on high-resolution aerial images.
- TweeTopics: Uncovering Patterns with BERT and LDA** | *Python, BERT, LDA* 2022
- Implemented BERT and LDA for deep learning text and topic modeling, employing unsupervised learning algorithms to analyze social discourse dynamics on Twitter.
- Urban Pedals: Visualizing BikeShare Dynamics in Chicago** | *Python, R, Tableau* 2021
- Utilized Pandas, R for data analysis, and Tableau, JavaScript for visualizations, applying clustering algorithms and time-series analysis to uncover patterns in Chicago’s BikeShare system, revealing urban transportation patterns.
- Predicting popularity using Altmetric** | *Python, ScikitLearn, API (Altmetric)* 2020
- Employed ScikitLearn and Python for predictive modeling using regression algorithms and decision trees to forecast future citation counts of scholarly articles using Altmetric data, identifying early impactful research.

INVITED TALKS & PRESENTATIONS

- **Fluid Multi-View Analytics: Generative Multi View Systems**, *DePaul University CDM*, Chicago, 2025.
- **iTrace: Interactive Tracing of Cross-View Data Relationships**, *University of British Columbia*, BC, 2025.
- **MSW Predictive Modeling: A comprehensive approach**, *CEET Innovation Showcase*, DeKalb, 2024
- **Multi-Class Building Classification**, *IIN Sustainability Research Conference*, Chicago, 2024
- **Utilizing Multiple Views to analyze data**, *Northern Illinois University, CSCI 658*, DeKalb, 2023.

- **Toward Systematic Design Considerations of Organizing Multiple Views**, *IEEE VIS* conference, 2022
- **Exploring COVID data using Instagram Images**, *Northern Illinois University, CSCI 656*, DeKalb, 2022.
- **Modeling the Broader Impact of Science**, *Northern Illinois University, CSCI 600*, DeKalb, 2022.
- **Organizing layout of Multiple View systems**, *Northern Illinois University, CSCI 628*, DeKalb, 2021.
- **Cross-Device interaction to explore 3D Vis**, *Northern Illinois University, CSCI 626*, DeKalb, 2021.
- **Predicting Patent Citations to measure Economic Impact**, *JCDL 2019*, Urbana-Champaign, 2019.
- **Visualizing BikeShare Dynamics in Chicago**, *Northern Illinois University, CSCI 627*, DeKalb, 2019.

HONORS & AWARDS

- **Best Paper Honorable Mention**, *IEEE VIS*, 2022
- **Society Involvement Award**, *Honor Society*, 2020
- **Community Service Award**, DeKalb, 2019

LEADERSHIP

- **Lab Head**, *Visual Analytics Lab*, NIU, DeKalb, 2020 – present
- **Mentor**, *WASTE Lab*, NIU, DeKalb, 2024 – 2025
- **Chief Coordinator**, *Computer Society of MJCET*, Hyderabad, 2017
- **Leader – Web/Graphic Design Head**, *Entrepreneurship Cell, MJCET*, 2016
- **Lead Developer**, College's Annual Technical Fest Website (Adsophos), 2015

SERVICES

Program Committee Member

- CIKM 2026, CIKM 2025, CIKM 2024, CIKM 2023
- WWW 2026
- JCDL 2025

Reviewer

- Scientometrics Journal
- Frontier of Psychology
- New Media & Society
- PacificVis
- WebSci
- PLOS ONE

Professional Memberships

- IEEE
- ACM

SKILLS

Programming: Python, Java, C/C++, JavaScript/TypeScript, SQL, R, HTML/CSS, LaTeX

ML & CV: PyTorch, TensorFlow/Keras, Hugging Face, Scikit-learn, OpenCV, ViT, CLIP, SAM

GenAI & LLM: RAG, prompt engineering, tool calling, embeddings, LoRA/QLoRA, APIs, LLaMA, Stable Diffusion

Agents & Orchestration: LangChain, LangGraph, AutoGen, CrewAI, ReAct, n8n, Flowise, LlamaIndex

Retrieval & Data: FAISS, ANN indexing, hybrid search (BM25+dense), reranking, PostgreSQL, MongoDB, Kafka

Evaluation & Observability: A/B testing, SHAP/LIME, hallucination/bias testing, Ragas/TruLens

MLOps & Deployment: Docker, Kubernetes, FastAPI, MLflow, Weights & Biases, DVC, Git/GitHub Actions, CI/CD

Visualization: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Streamlit/Dash, Tableau/Power BI, D3.js